

IMPLEMENTING OBSERVABILITY OF NJ CONTACT CENTER HEALTH

New Jersey Innovation Authority | Contact Center Modernization

Will-Josh Njeumi, Software Engineering Fellow | *University of Maryland, Information Science*

Keywords:

infrastructure, observability, monitoring, notification

Summary:

To provide observability and system monitoring for New Jersey's contact centers, Will-Josh collaborated with the NJIA's CCM team to create a statewide notification system to proactively identify broken user experiences and contact center outages. Utilizing **Infrastructure as Code** (IaC) via **TypeScript**, he leveraged the **AWS CDK** framework to implement alarms that notify the team when a failure in the system occurs. This moves the state from fifteen-plus agencies each operating in the dark about their own systems to comprehensive dashboards and automated alerts — and lays the groundwork for understanding not just when systems break, but how well the system as a whole is serving residents.

IMPLEMENTING OBSERVABILITY OF NJ CONTACT CENTER HEALTH

Resident Experience
Contact Center Modernization
James Viall — Lead Software Engineer

coding it forward >



New Jersey
Innovation Authority

WILL-JOSH NJEUMI
University of Maryland
Information Science

CONTACT CENTER MODERNIZATION PROJECT

MISSION AND STRATEGY

- Works with 15+ State agencies to streamline and modernize contact center operations by transitioning legacy contact centers that require manual operations to cloud-based systems (Amazon Web Services/AWS)

6-15%

Increase in number
of callers supported
(talked to an agent)

40-60%

Reduction in costs.
Average wait times
reduced to 1-2
minutes

50%+

Increase in callers
that self-serve

\$1-2m

A year in savings
and additional
capacity

SCALE

- Varying levels of load and scale across each contact center, with Treasury, Temporary Disability, and Unemployment Insurance receiving **thousands of calls per day**
- Providing features across phone, live chat, and SMS channels reduces wait times for callers, makes it easier for agency stakeholders to manage and maintain their contact centers, and improves user experience end-to-end for supervisors, agents, and residents alike

THE PROBLEM

BACKGROUND

- Investigating issues often took **an hour or more** due to having to **manually search** through logs and correlate issues across pieces of the platform
- Deployments require **manual intervention** from an engineer, sometimes taking **an entire day** to complete
- Diagnoses happened **retroactively** and team lacked capacity to check **all** contact centers regularly
- No way to see the health of a state contact center via a single view
 - Had to look in many different places across the AWS console in order to observe the health of every aspect of the system (Connect, Lambdas, databases, etc.)
- Ability to diagnose issues was mostly independent from one agency to the next, with little visibility into whether the same problem was happening elsewhere

Solution: Statewide Contact
Center **visibility and outage
detection**

OBSERVABILITY & MONITORING

- Created CloudWatch alarms to track health of CCM resources (AWS Lambdas, Connect service)
- Alarms were placed under one “composite” alarm that gives a summary of overall system health
- Sends a notification if at least one of the alarms goes into a red, “In Alarm” state
- Integrated alarms into team’s system health dashboard for quick glance view of issues

COMPOSITE ALARM

The screenshot displays the AWS CloudWatch Alarms console. The main view shows the 'njooi-ccm-nj-treasury-SystemHealth' composite alarm, which is currently in an 'OK' state. A timeline graph shows the alarm's state changes over the last 15 minutes. Below the timeline, a table lists the child alarms, all of which are also in an 'OK' state.

CloudWatch Alarms (77)

- Alarms: 43 OK, 2 In alarm, 32 Muted
- Alarms: **Alarms** (43 OK, 2 In alarm, 32 Muted)
- Al Operations
- GenAI Observability
- Application Signals (APM)
- Infrastructure Monitoring
- Logs
- Metrics
- Network Monitoring

njooi-ccm-nj-treasury-SystemHealth

Timeline: This shows the state changes of the composite alarm.

Child alarms:

Name	State	Last state update	Type	Alarm rule
njooi-ccm-nj-treasury-SystemHealth	OK	2026-08-11 08:...	Metric alarm	-
njooi-ccm-nj-treasury-SystemHealth	OK	2026-08-11 08:...	Metric alarm	-
njooi-ccm-nj-treasury-SystemHealth	OK	2026-08-11 08:...	Metric alarm	-
njooi-ccm-nj-treasury-SystemHealth	OK	2026-08-11 08:...	Metric alarm	-
njooi-ccm-nj-treasury-SystemHealth	OK	2026-08-11 08:...	Metric alarm	-
njooi-ccm-nj-treasury-SystemHealth	OK	2026-08-11 07:...	Metric alarm	-

njooi-ccm-nj-treasury-CallbackAttemptMissedAlarm

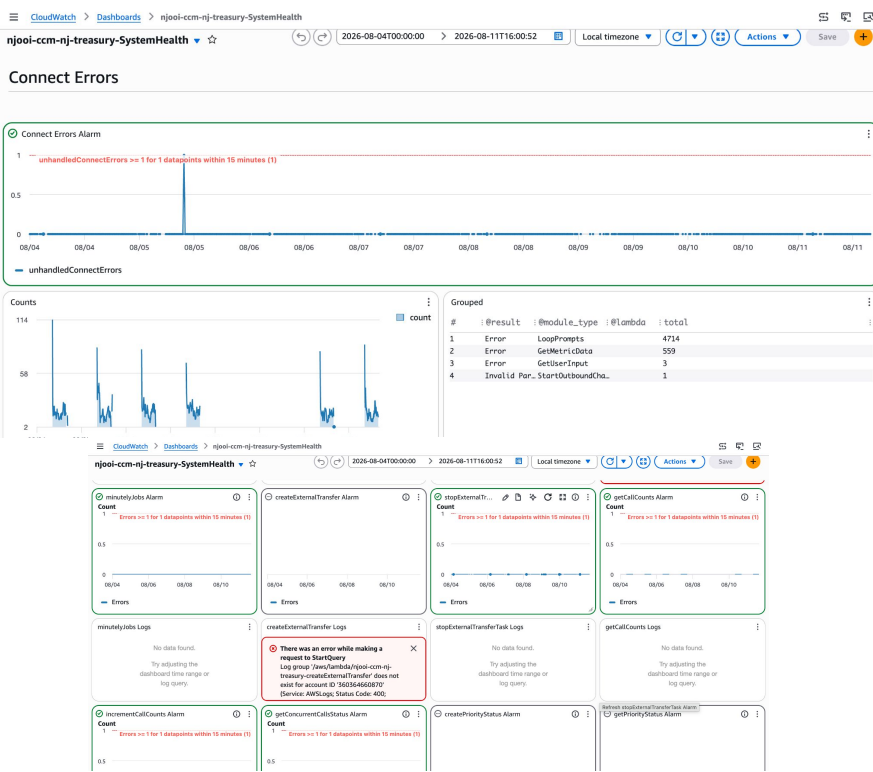
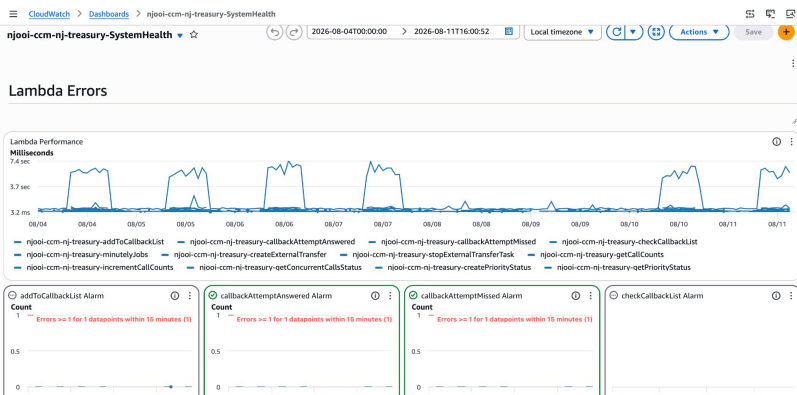
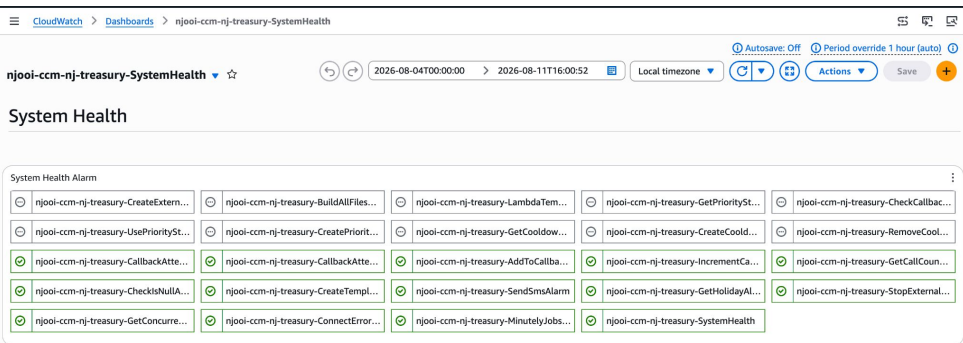
Details:

Name	State	Namespace	Datapoints to alarm
njooi-ccm-nj-treasury-CallbackAttemptMissedAlarm	OK	AWS/Lambda	1 out of 1

Actions Actions enabled

Type	Description
Notification	When alarm transitions to in alarm, send message to topic "njooi-ccm-nj-treasury-SystemHealthNotifications".

DASHBOARD



TRIAGING ISSUES



**CloudWatch Alarm** 10:53 AM
Email ▾

 **ALARM: "njooi-ccm-appeals-ui-SystemHealth" in US East (N. Virginia)**
Aug 7th

 1  1 

 1 reply 3 days ago

**CloudWatch Alarm** 11:11 AM
Email ▾

 **ALARM: "njooi-ccm-aws-dol-wage-and-hour-call-center-SystemHealth" in ...**
Aug 7th

 1  1 

 2 replies Last reply today at 7:58 AM

**CloudWatch Alarm** 11:31 AM
Email ▾

 **ALARM: "njooi-ccm-nj-working-papers-SystemHealth" in US East (N. Virgi...**
Aug 7th

 1 

**Thread** 

**CloudWatch Alarm** Jul 23rd at 5:12 PM
Email ▾

 **ALARM: "njooi-ccm-nj-treasury-sum-c..."**
Jul 23rd

 1 

7 replies

**Lynn Herrick** Jul 23rd at 6:23 PM
us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#a.id=%7E%278d1abe51-9009-...&a.pos=%7E%2706&...

This was a timeout on an interactive message for yes/no/agent at the within lambda.

**Lynn Herrick** Jul 23rd at 6:31 PM
Duration billed was 8000.00 ms. This is outside of our expected behavior for Interactive Messages and this particular Lambda. (edited)

Screenshot 2026-07-23 at 3:30.42 PM.png ▾



 2 

**Lynn Herrick** Jul 23rd at 6:37 PM
My next hunch is that any of the api calls within this lambda caused issue. Connect Data Tables 2 times within a single lambda call. There are currently no debug logs around those calls and we may want to add them if we continue to see issue here.

It looks like (I cannot find) there are no pre-made metrics for Connect Data Tables. Which means I can't track the latency there to overlay with this graph.

**Lynn Herrick** Jul 23rd at 6:50 PM

VALUE

- Surfaced a spike in an error breaking a self-service flow for a small number of callers
- Same error was visible across multiple call center instances at once
- Team immediately recognized it as a platform-level AWS anomaly rather than an isolated local issue
- Was able to immediately escalate the issue to AWS and take action to diagnose the root cause



Caroline Schauder 11:49 AM

@James Viall we're getting pretty consistent errors in lottery for the [Show View](#) module in [View Flow](#) . I wrote up some stuff about it in one of the threads before, but was wondering if you know anything about this, and if its something we need to investigate more?



6 replies Last reply 18 days ago

IMPACT

- Awareness of issues are no longer retrospective and, at times, days late
- Alerts informs team about issues within **15 minutes** of them happening
- Investigation time substantially reduced
- Greatly increased confidence in deployments, requiring much less manual oversight, enabling team to fully automate deployments without manual intervention in the near future
- NJIA now has a strong foundation of measuring the health of a contact center, setting the stage for:
 - More sophisticated alarms targeting issues around seasonality, load, and other anomalies
 - Automated rollbacks when issues occur after a deployment
 - Unified view of all statewide contact centers

TAKEAWAYS

- First time working on infrastructure and seeing how it differs from frontend, backend, and data
 - Can confidently say I enjoy backend and infrastructure work the most, aspire to be in roles of those focuses in the future
- Glimpse into what it's like working in state government
- Got a behind the scenes look of a modern state call center and what goes into building, maintaining, and improving them
- Felt gratified hearing how beneficial the changes I made were to the team and how much easier it'll make their work

THANK YOU!

Special shoutouts to James and the CCM team, Cassie, CIF, and my NJIA fellows! (Kapila, Stephen, Nate)